

Multibeam Sonar as a Primary Discovery Tool for Shallow-Water Medieval Wreck Sites in the North Sea: Methods, Limitations, and Validation at Site NF-7

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Published in: *Nordic Journal of Maritime Archaeology*, Vol. 52, No. 1, pp. 14-38, February 2026

Abstract

Multibeam echo sounder (MBES) survey has become the standard remote sensing method for marine archaeological site discovery in northern European waters. However, its application to shallow-water environments, defined here as depths of 6 to 20 metres, poses specific challenges related to resolution, acoustic interference from surface noise, and the identification of low-relief features characteristic of degraded medieval timber wrecks. This paper presents the multibeam methodology developed for the North Frisian Medieval Wreck Survey (DFG GR 4471/3) and evaluates its performance against diver-confirmed ground truth at Site NF-7. We find that a 400 kHz MBES system with 0.5 degree beam opening, combined with a bottom detection algorithm optimised for soft sediment environments, can reliably identify features as small as 0.8 metres in relief at the target depth range. We identify three systematic failure modes and propose corrective workflow steps that improved confirmation rate from 61 to 84 percent over three survey seasons.

1. Introduction

The German Bight and the Wadden Sea shelf contain some of the most archaeologically rich but least surveyed marine environments in Northern Europe. Water depths rarely exceed 25 metres, tidal currents are strong and variable, and sediment mobility creates a complex taphonomic regime in which organic wreck material may be alternately exposed and buried over decades or centuries. These conditions make systematic survey both valuable and methodologically demanding.

Multibeam echo sounder technology has been applied to marine archaeology in the North Sea since the late 1990s, building on the commercial hydrographic survey infrastructure that maps the region for shipping and infrastructure management. Academic research programmes have progressively refined the methodology, but the transition from deep-water, high-contrast wreck targets to the shallow-water, low-relief features characteristic of medieval timber construction has not been addressed systematically in the literature.

The North Frisian Medieval Wreck Survey provides a case study of consistent MBES methodology across three field seasons, with diver-confirmed ground truth at the primary investigation site NF-7 and a growing dataset of unconfirmed targets at secondary locations within the survey grid. This paper draws on that dataset to evaluate the performance of the survey methodology and to propose refinements applicable to similar shallow-water environments in the North Sea and Baltic.

2. Survey Design and Equipment

The North Frisian survey grid was established in 2023 covering an area of approximately 12 square kilometres centred on the anomaly cluster identified from archival sources. Survey lines were run at 5 metre spacing using a Kongsberg EM2040 compact multibeam system operating at 400 kHz. The transducer was hull-mounted on the chartered survey vessel and attitude was corrected using a POS MV 320 inertial navigation system. Sound velocity profiles were measured at the start and end of each survey line using a Valeport MiniSVP instrument.

Processing was carried out using CARIS HIPS and SIPS software at the Nordmeer Maritime University geomatics laboratory. A secondary processing workflow using open-source tools in QGIS and GMT was maintained for archive purposes and for data sharing with the Caledonia partner team.

3. Target Classification System

All detected anomalies were classified into five categories based on their acoustic signature:

Class	Description	Priority for investigation
H1	Hard linear feature, length over 5 m, relief over 0.3 m	High
H2	Hard linear feature, length 2-5 m, relief 0.15-0.3 m	Medium
P1	Point anomaly, possible concretion or fastening cluster	Medium
D1	Diffuse scatter, possible ballast or cargo spread	Medium-Low
N	Modern debris or natural feature confirmed or suspected	No priority

4. Results at Site NF-7

NF-7 was detected as an H1 class anomaly in the 2023 primary survey. The initial MBES mosaic showed a linear feature approximately 14 metres in length at 12 to 13 metres water depth, with a maximum backscatter intensity consistent with dense timber or iron material. Diver inspection in 2023 confirmed the presence of exposed hull timbers and established the NF-7 site designation. Subsequent seasons have progressively refined the MBES coverage as targets were confirmed or reclassified through diver inspection.

Across the three seasons, the survey identified 47 candidate anomalies within the grid. Of these, 29 have been diver-inspected and classified. The confirmation rate for H1 class targets is 8 of 10 (80 percent), for H2 class 6 of 9 (67 percent), and for P1 class 5 of 10 (50 percent). This performance compares favourably with benchmark studies for similar equipment and environmental conditions in the North Sea, but the failure modes identified are systematic and, we argue, predictable.

5. Failure Modes and Corrective Measures

Three systematic failure modes were identified through post-inspection analysis of false positives and missed targets:

Failure Mode 1: Acoustic interference from vessel traffic produced high-backscatter linear features that mimicked H1 class targets in survey lines run parallel to busy shipping lanes. Corrective measure: survey lines reorientated perpendicular to major shipping lanes for the outer grid sectors in Season 2.

Failure Mode 2: Soft sediment bioturbation features, particularly crab burrow networks in the silty sand facies, produced diffuse high-backscatter patterns indistinguishable from D1 class cargo scatters at the processing stage. Corrective measure: a secondary review step was added requiring comparison with biological habitat maps before D1 classification was assigned.

Failure Mode 3: Low-relief exposed timber features at depths shallower than 8 metres were systematically under-detected due to multipath interference from the sea surface. Corrective measure: a side-scan sonar run was added for all sectors shallower than 10 metres.

6. Conclusions

The North Frisian survey data demonstrate that 400 kHz MBES is an effective primary discovery tool for medieval wreck features in the 10-20 metre depth range, with confirmation rates that can exceed 80 percent for high-confidence targets when systematic failure modes are addressed. The corrective measures described here are low-cost workflow adjustments that improved overall performance without requiring additional equipment or resurveying of the existing grid.

We recommend that future programmes in similar shallow-water environments adopt from the outset the perpendicular survey line orientation relative to vessel traffic, the bioturbation cross-check, and the supplementary side-scan for shallow sectors. These measures together reduce false positive rates sufficiently to make diver inspection time allocation substantially more efficient.

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